Owning the Chokepoint: ASML, Alliance Dependence, and Europe's Leverage in the AI Race

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Abstract

Commentators call ASML, the only maker of the extreme ultraviolet lithography machines on which advanced AI chips depend, Europe's last card in the US–China race for artificial intelligence. Yet the Netherlands has aligned its export controls with American demands since 2019 with little to show for it, and a widely read scenario, *Europe 2031*, imagines Europe surrendering control of the firm. We develop a game-theoretic model in which a security-dependent ally holds a chokepoint its patron needs while the patron's rival bids for access. The holder's leverage equals its exit premium: the rival's credible offer net of the costs the patron can credibly impose. The patron's dependence on the chokepoint matters only through sanctions that would harm the patron itself, the holder's agenda power, the risk of breakdown, and the holder's power to hurt during a standoff. Defence dependence gives the patron leverage only insofar as its guarantee is conditional. The rival's coercion raises the patron's price without helping the holder. And an anticipated rise in the holder's exit premium gives the patron a preventive motive to demand control of the chokepoint while its own threats still bite. Applied to the scenario and to the record since 2019, the model accounts for Dutch compliance without compensation, finds that the scenario's 2029 Council vote cannot fail as described, and identifies the conditions under which ASML would become a card worth holding.

*Working draft, October 28, 2026. Replication code for all numerical checks and figures is in the project repository.

1 Introduction

On June 11, 2026, nine European researchers, investors, and policy writers published *Europe 2031*, a scenario of how the continent loses its place in the age of artificial intelligence (**?**). Its climax is a phone call. In March 2031 the US vice president tells the leaders of the Netherlands, Germany, and France that ASML, the Dutch firm that alone builds the extreme ultraviolet (EUV) lithography machines through which every leading-edge AI chip passes, is to be folded into a holding company in which Washington holds the controlling vote over output, customers, and technology transfer. The authors call ASML "the only card Europe still holds" and "the one bottleneck the entire AI race runs through". Yet in their story Europe never plays it. In 2028 the Dutch halt their remaining sales of older lithography tools to China under the threat of US secondary sanctions, and Europe "negotiates nothing in return". In 2029 an EU countermeasure aimed at ASML's deliveries to American fabs fails in the Council. In 2031 Europe is left choosing among submission to Washington, alignment with Beijing, and isolation.

The puzzle is not confined to fiction. Since 2019 the Netherlands has aligned its export controls with American demands in a series of steps: it declined to renew ASML's licence to ship an EUV machine to China in 2019, imposed national licensing on advanced immersion lithography in 2023, revoked part of a licence in January 2024, and extended its controls to older tools, spare parts, and software in September 2024 (**?**). The main thing The Hague received in return was an exemption, in December 2024, from American rules that would otherwise have applied to the same tools, and Washington can withdraw it with a notice in the Federal Register (**?**). When Congress moved in 2026 to impose a countrywide ban on servicing China's installed base of Dutch tools, the Dutch trade minister protested that the bill would "put our national security in American hands" (**?**). If ASML is the chokepoint of the most consequential technology race of the century, why does its owner seem to extract so little for it?

The literatures that should answer this question point the other way. Theories of asymmetric interdependence hold that the less dependent party to an exchange gains influence over the more dependent one (**??**). Work on weaponized interdependence argues that states with jurisdiction over central nodes in global networks can turn them into instruments of coercion (**?**). Both imply that the United States, which cannot extend its lead in AI without ASML's machines, should be paying Europe for continued access. Formal models of sanctions and issue linkage analyze a sender and a target (**??**). Models of alliance coercion study a patron pressing a dependent client (**??**). Recent work in geoeconomics locates a hegemon's power in its ability to coordinate threats across relationships (**?**). None of these considers a dependent ally that holds a chokepoint its patron needs, in a race in which a rival great power would pay for access.

We develop a game-theoretic model of that situation. Europe controls a chokepoint and decides whether to keep it aligned with its security patron, the United States, or to let it serve the patron's rival, China. Both great powers make offers, and each controls instruments linked to Europe's choice: security protection and access to frontier AI on the American side, critical inputs on the Chinese side. A threat to use any of these instruments counts only if carrying it out would be sequentially

rational.

We argue that a chokepoint is worth what its holder could credibly gain by walking away, not what its patron would lose without it. We call the first quantity Europe's *exit premium*: the most China can credibly offer, net of the costs the United States can credibly impose and the value Europe itself places on alignment. When the exit premium is negative, Europe receives nothing however much Washington depends on ASML. The patron's dependence returns only through four channels: sanctions whose use would harm the patron itself, the holder's ability to commit to a demand, the risk that a dispute slips out of control, and the holder's ability to inflict pain while bargaining continues.

The model yields five sets of results. First, leverage equals the positive part of the exit premium, and a positive premium is necessary for any leverage under every bargaining protocol we consider. Second, Europe's dependence on American protection gives Washington leverage only to the extent that the guarantee is *conditional*, reliable for a loyal Europe and void for a realigned one. Compute scarcity first creates an American AI lever and then destroys it, once Washington rations loyal allies too. China's capacity to coerce Europe raises what Washington must pay for European alignment but never raises Europe's own payoff, while China's promises help Europe only when it already has leverage. Third, because Europe cannot commit not to exploit a rising exit premium, Washington has a preventive motive to demand control of the chokepoint while its own threats still bite. It needs a public commitment to do so, and a Chinese threat to punish the transfer of control protects European ownership. Fourth, Europe's chokepoint is nested in American ones. ASML depends on US-origin subsystems, US staff, and US customers, so Europe's exit threat is only as credible as the Dutch state's willingness to compel its own champion, and Washington needs to win over only the cheapest veto player. Fifth, the one place where American dependence on ASML matters directly is a war of attrition over servicing. There, what decides the outcome is time to pain: the length of the US stockpile relative to Europe's endurance.

We use the model to reread both the scenario and the record of 2019–2026. The 2028 capitulation in *Europe 2031* is an equilibrium of the game the authors describe. The 2029 vote is not, because the states the scenario names cannot form a blocking minority under 2026 Council population weights. The 2031 ultimatum is a preventive demand for control rights, but its threat of permanent exclusion from American AI is not credible. In the real record, Dutch alignment without compensation, the December 2024 US exemption of allies with equivalent controls, the Nexperia dispute of 2025, and the 2026 fight over servicing restrictions are what the model predicts for a holder whose exit premium is negative.

The paper contributes to four literatures. First, it qualifies the theory of weaponized interdependence by showing that control of a hub yields leverage over the hub's main user only through a credible alternative, and by formalizing the institutional condition that a nested chokepoint fails. Second, it separates two things the alliance literature often treats together, an ally's dependence on its patron and the conditionality of the patron's commitment, and shows that only the second is a source of the patron's leverage. Third, it extends the analysis of commitment problems under

shifting power from preventive war to preventive demands for control rights over an asset that shapes future bargaining power (**???**). Fourth, it gives the European debate about technological sovereignty a set of conditions under which ASML is, and is not, a card worth holding.

Section 2 describes ASML's position and the three dependencies that shape Europe's use of it. Section 3 sets out the arguments we build on and against. Section 4 presents the model and its main results, section 5 the analysis of control rights, and section 6 three extensions. Section 7 applies the results to the scenario and to the record since 2019. Section 8 draws out implications and limitations.

2 The ASML Question

2.1 A chokepoint

Every chip made at the leading edge, including the accelerators and high-bandwidth memory on which frontier AI runs, is patterned with EUV lithography, and ASML is the only firm that makes EUV machines. It recognized 48 EUV systems in 2025, four of them the newer High-NA version, and expected to ship about 65 low-NA systems in 2026. In October 2026 its chief executive said that planned supply for both 2027 and 2028 would not meet customer demand (**??**). The machines cannot simply be bought and left alone. Installed-base management, ASML's term for servicing, upgrades, and spare parts, made up a quarter of its 2025 revenue, and EUV tools reportedly stop working quickly without ASML parts (**??**).

China cannot buy EUV machines, but it is ASML's largest market for the older immersion DUV tools with which Chinese fabs make 7-nanometre chips by multiple patterning. Customers in China accounted for 26 percent of ASML's sales in 2023, 36 percent in 2024, and 29 percent in 2025 (**?**). ASML expects China to fall to about 20 percent in 2026, although Chinese memory makers pulled orders forward in the third quarter, reportedly to stockpile tools against a future cutoff (**??**). Several hundred Dutch immersion tools are believed to be running in China, and they require maintenance roughly every six months (**?**). China's own efforts to escape the chokepoint are real but slow. A prototype EUV light source built in Shenzhen with former ASML engineers had not produced working chips by the end of 2025, and Beijing's target of 2028 for working chips is widely considered optimistic (**?**).

2.2 A nested chokepoint

ASML is a Dutch company, but the machine it sells is not a Dutch product in any simple sense. Carl Zeiss SMT in Germany is the sole supplier of its optics, and ASML says that a prolonged loss of Zeiss supply would "effectively" end its business. The EUV light source is developed at Cymer in San Diego and driven by laser amplifiers from Trumpf. At the end of 2025 ASML had 8,242 employees in the United States, about a fifth of its workforce, and its manufacturing sites include Wilton, Connecticut, and San Diego (**?**). This geography gives Washington jurisdiction over the firm.

When the United States restricted support for advanced Chinese fabs by US persons in October 2022, ASML instructed its American staff to stop servicing Chinese customers within a week (**?**).

The United States has turned that jurisdiction into a standing instrument. A December 2024 rule extended the foreign direct product rule to semiconductor manufacturing equipment and set a zero de minimis threshold for certain lithography tools bound for China. It exempted countries with equivalent national controls, among them the Netherlands, Japan, and most EU member states, through a supplement that the United States can amend at will (**?**). In April 2026 a bipartisan bill, the MATCH Act, proposed a countrywide ban on immersion lithography and on servicing for listed Chinese fabs, with 150 days for allies to align before Washington would impose controls of its own (**?**). In the terms of **?**, ASML satisfies the network condition for weaponization but not the institutional one: the state that hosts the node does not control the networks in which the node is embedded.

2.3 Three dependencies

Europe's use of ASML is shaped by three dependencies, each of which the model treats explicitly.

Security. Europe relies on the United States for nuclear deterrence, air and missile defence, intelligence, and enablers it cannot quickly replace. Bruegel and the Kiel Institute estimate that defending Europe without the United States would require about 300,000 additional troops and about EUR 250 billion a year in additional spending (**?**). That sum is roughly eight times ASML's 2025 revenue and about 25 times its sales to China. The second Trump administration has made the link between protection and economic alignment explicit. Its December 2025 National Security Strategy promises "more favorable treatment on commercial matters, technology sharing, and defense procurement" to partners that "align their export controls with ours" (**?**). It has also cast doubt on the guarantee itself. Withdrawals from Germany were announced in 2026 and a larger drawdown was under review that September, and in a May 2026 poll majorities in all fifteen European countries surveyed doubted the United States would come to their defence (**??**).

Frontier AI. Europe hosts about five percent of the world's AI compute and depends on American firms for frontier models (**?**). Access has already been treated as an instrument of policy. The January 2025 AI Diffusion Rule divided the world into tiers and placed ten EU members in the top tier and the other seventeen below it (**?**). The August 2025 joint statement on the EU–US trade framework tied European purchases of US AI chips to European adoption of US technology-security requirements (**?**).

China. Europe imports almost all of its rare-earth permanent magnets from China, and Beijing has shown that it can use licensing to cut supply in weeks. Its October 2025 controls, which reached foreign products containing small amounts of Chinese rare earths, were suspended after the US–China meeting in Busan, not withdrawn (**?**). The Nexperia dispute is the closest real analogue

to the scenario's endgame. On September 30, 2025, the Dutch government used emergency powers to take control of the Chinese-owned chipmaker after a US rule threatened its operations. China banned exports from Nexperia's Chinese plant, European carmakers warned of stoppages within days, and on November 19 the Dutch minister suspended the order (**?**). Relief came mainly through the US–China deal, not through anything Europe obtained for itself.

2.4 What the scenario claims

Europe 2031 makes five claims that a model can test. (1) In 2028, faced with US secondary sanctions, the Netherlands has no choice but to halt servicing of Chinese DUV tools, and gets nothing for it. (2) In 2029 Europe cannot use ASML against the United States because security-dependent members and the Netherlands itself block the countermeasure. (3) In 2031 Washington moves to take control of ASML because it fears losing the chokepoint to a Europe drifting toward China. (4) Its threat of permanent exclusion from US AI, backed by a stockpile of EUV tools, lets it outlast Europe. (5) Compute on European soil and a coalition of middle powers would have given Europe leverage. The model confirms (1) and (3), rejects the mechanism behind (2), finds the threat in (4) incredible as stated, and specifies when (5) would and would not work.

3 Leverage, Linkage, and Commitment

Three bodies of work bear on the ASML question. Each supplies part of the model, and each, applied on its own, gives the wrong answer about where Europe stands.

3.1 Interdependence and chokepoints

? argued that a state gains influence over trading partners who would lose more than it would from ending their exchange. **?** refined the idea by distinguishing sensitivity, the costs a cutoff imposes before anyone adjusts, from vulnerability, the costs that remain after adjustment. Work on weaponized interdependence moved the argument from bilateral trade to networks. States with jurisdiction over hubs in hub-and-spoke networks can exploit them to gather information or deny access, provided their domestic institutions let them do so (**???**). Applied to the semiconductor supply chain, this logic has been used to explain American power over China (**??**) and, by extension, to suggest that the holders of other chokepoints, including the Netherlands, hold cards of their own.

? identified the problem with this inference. Asymmetric interdependence implies nothing about political influence unless trading economic for political concessions leaves both sides better off than bargaining over the economic surplus alone. The less dependent state gains influence only through credible threats that change the other side's disagreement payoff. **?** makes a related point about exit costs. Our model takes Wagner's position and makes it precise for a three-player setting. The holder of a chokepoint has leverage over its main user only if it has a credible alternative, and the alternative is set by the rival's willingness to pay net of the costs the main user can impose. Recent

work also treats chokepoints as depleting assets: activation induces rerouting and indigenization (**??**). That decay is one reason the timing of leverage matters in section 5.

3.2 Linkage and alliance coercion

The formal literature on sanctions treats coercion as bargaining in which outcomes depend on the parties' relative toughness and on what each would suffer (**?**). Because targets that would rather concede do so before sanctions are imposed, successful coercion is usually unobserved (**??**). Coercive cost, more than revealed resolve, appears to drive the success of threats (**?**). Coercers who link issues can trade concessions across them (**??**). A hegemon that can coordinate threats across many economic relationships gains power that no single relationship would give it (**?**).

These models typically pit a sender against an adversary. In the ASML case the sender would be a junior ally that depends on its target for protection. Alliance theory has long held that asymmetric alliances trade the minor partner's autonomy for security (**?**), that allies fear abandonment and entrapment (**?**), and that patrons can exercise authority over subordinates in exchange for protection (**?**). The United States has used that authority before. Conditional threats of abandonment helped stop West German nuclear ambitions (**?**). Toshiba's 1987 sale of milling machines to the Soviet Union led to US sanctions on the firm (**?**). Coercion of this kind needs an assurance that compliance will be rewarded as well as a threat, and the two can work against each other (**??**). The record also contains a counter-example. In 1982 Washington tried to stop European firms from supplying the Soviet gas pipeline through extraterritorial sanctions and backed down within months once Britain, France, Italy, and West Germany stood behind their firms (**??**).

Our model brings these strands together. The same dependence that allows the patron to coerce the holder also lets it neutralize the holder's chokepoint. Whether it does so depends on the conditionality of its commitments, not on their size, and on whether carrying out each threat would hurt the patron. Firm-level penalties that would disrupt the patron's own supply are where **?** and the 1982 episode belong. Jurisdiction over flows bound for the rival is a different and much cheaper instrument.

3.3 Rivals, bidders, and wedges

The third player in the ASML question is China, which would gain from Europe's realignment and can both reward and punish. The literature on wedge strategies studies how a state detaches members of a hostile coalition by rewards or by coercion (**??**). Formal work on third parties in bargaining shows that a prospective ally may signal its alignment by taking a hard line with its patron's rivals (**?**). And when two outside powers compete for a client's policy, the client's price depends on the competition between them (**?**). The closest formal structure to our baseline is sequential vote buying, in which a first mover must make every winning coalition too expensive for a second mover (**?**), and the related theory of menu auctions (**?**). In both, the winning bidder pays roughly the loser's valuation. Our model adds what these frameworks lack for the ASML case: one

bidder also controls the client's security and the jurisdiction under which the client's firm operates, and the credibility of each instrument is derived rather than assumed.

3.4 Commitment and control

The last ingredient is time. **?** and **?** show that anticipated shifts in bargaining power create commitment problems that can make bargaining fail even under complete information. **?** considers bargaining over objects that affect future bargaining power, and **?** study preventive action against latent capabilities. Control of ASML is such an object, and the scenario's Pentagon compares losing it to nuclear proliferation. The theory of the firm offers the relevant concept: ownership confers residual control rights, which determine how future surplus is divided when contracts are incomplete (**?**). Section 5 combines the two. A patron that anticipates the holder's exit premium will rise can buy or coerce control rights while its threats still bite, and the holder's sovereignty costs and the rival's threats determine the price.

4 A Model of Chokepoint Bargaining

The model has three players: Europe (E), the United States (U), and China (C). Europe controls whether ASML's tools, spare parts, and service engineers reach Chinese fabs. In practice that control is split among the Dutch licensing authority, the firm, and the European Union, and we take the split up in Section 6.1. Until then we treat Europe as a unitary actor. The model is built to answer one question: what determines the price Europe can charge the United States for keeping ASML on the Atlantic side of the AI race?

4.1 Setup

Alignment. Europe chooses an alignment $a \in \{A, X\}$. Under Atlantic alignment (A), ASML supplies only US-aligned fabs and complies with every US denial requirement. Under exit (X), Europe lets ASML supply China with tools, servicing, or licences that the United States wants withheld. Exit stands for any realignment that moves frontier lithography toward China. Section 6.3 adds an intermediate option, hedging, which corresponds to the legacy immersion DUV sales and servicing at issue in the 2023–2025 Dutch export controls and in the scenario's 2028 crisis.

Stakes. The United States loses $V > 0$ if Europe exits. V measures the patron's dependence on the chokepoint: what America gives up in the AI race if China gains access to ASML's tools. China gains $W > 0$ from Europe's exit. A share $\theta \in [0, 1]$ of that gain can be blocked by US jurisdiction over China-bound flows, through the foreign direct product rule, de minimis rules for tools containing US-origin content, and control of US-made subsystems. Enforcing that jurisdiction against China-bound shipments costs the United States little, so we treat it as credible throughout and let θ capture its reach.

Promises. Each great power can offer Europe a transfer conditional on alignment. The US offer $\tau_U \geq 0$ is paid if Europe chooses A; China's offer $\tau_C \geq 0$ is paid if Europe chooses X. Transfers are in kind: tier status under an AI export regime, exemptions from extraterritorial rules, tariff relief, market access, credit, or co-production. Promises must be self-enforcing, so we cap them at B_U and B_C, the largest transfers each power would still deliver after Europe has moved. Europe values a unit of China's in-kind offer at $\lambda_C \in (0, 1]$, which captures both the inefficiency of in-kind transfers and doubts about delivery. The most China can credibly put on the table, in units of European utility, is therefore

$$\omega \equiv \lambda_C \min\{(1-\theta)W,\, B_C\}. \tag{1}$$

Linked instruments. After Europe chooses its alignment, each great power decides how to use the instruments it controls. These decisions are moves in the game, so a threat matters only if carrying it out is sequentially rational. There are four instruments.

Security. With some probability a security crisis occurs in which Europe needs US protection. The United States defends Europe if its stake in doing so, s_a, exceeds a crisis-specific cost K drawn from a continuous distribution G. Europe loses $S > 0$ if it is left undefended. The probability of defence after alignment a is $\pi_a = G(s_a)$. We allow $s_A \geq s_X$: a Europe that supplies China's chipmakers may be worth less to Washington. The *conditionality* of the guarantee is

$$\kappa \equiv \pi_A - \pi_X = G(s_A) - G(s_X) \in [0, 1]. \tag{2}$$

Frontier AI. The United States can deny Europe access to frontier models and inference, which costs Europe $\alpha \geq 0$. Its ex post gain from denial after alignment a is $g_{ai}(a) = \nu - k_a$. Here ν is the opportunity value of the compute released, which is high when compute is scarce, and k_a is the value to the United States of continuing to supply Europe given its alignment, with $k_A > k_X \geq 0$. If the United States has publicly committed to deny frontier AI to states that realign, failing to apply that rule to an exiting Europe costs it $m_{ai} \geq 0$ in audience costs (**?**); supplying a loyal Europe violates no commitment.

Firm-level penalties. The United States can penalize ASML itself, by entity listing, denial of US-origin inputs, or withdrawal of US-person support, which costs Europe $f \geq 0$. Because ASML also supplies America's own fabs and those of its allies, penalizing the firm harms the United States. We write its ex post gain from penalizing an exiting Europe as $g_f = \gamma - \varphi V$, where γ is the added denial benefit and φV the self-inflicted cost. The associated audience cost is $m_f \geq 0$.

Chinese coercion. China can restrict exports of critical inputs, such as rare earths, magnets, or the legacy chips at issue in the 2025 Nexperia dispute, which costs Europe $r \geq 0$. Punishment reverses nothing in a one-shot game, so China carries it out only if its ex post value, g_r, which includes its reputation with other chokepoint holders, plus any audience cost m_r is nonnegative.

Values. Europe bears an intrinsic cost $\eta \geq 0$ from exit, reflecting the value it places on keeping frontier AI capabilities away from an authoritarian rival that supports Russia's war.

Timing. (1) The United States offers $\tau_U \in [0, B_U]$. (2) China observes τ_U and offers $\tau_C \in [0, B_C]$. (3) Europe chooses a. (4) Transfers are paid, nature determines whether a crisis occurs and draws K, and each power chooses how to use its instruments. Offers lie on a grid of mesh $\varepsilon > 0$ and we report limits as $\varepsilon \to 0$; Europe breaks ties in favour of A, and China does not bid when it would earn zero. The United States moves first because, in the scenario and in the 2023–2024 record, Washington sets the terms and others respond (cf. **?**). Section 4.4 shows which results depend on this choice.

Payoffs. We normalize payoffs relative to Atlantic alignment with no transfers. The United States receives $-\tau_U$ if Europe chooses A and $-V$ if Europe exits, where V is net of whatever the United States gains from its ex post responses. China receives $(1-\theta)W - \tau_C$ if Europe exits and zero otherwise. Europe's payoffs are stated in eq. (5) below, once we know which instruments are used.

4.2 Credible conditionality

Only the part of each instrument that depends on Europe's alignment can enter Europe's choice. The first step is to find which threats survive sequential rationality. Define the indicators

$$\chi_{ai} \equiv \mathbb{1}\{\nu + m_{ai} \geq k_X\}\mathbb{1}\{\nu < k_A\}, \qquad \chi_f \equiv \mathbb{1}\{\varphi V \leq \gamma + m_f\}, \qquad \chi_r \equiv \mathbb{1}\{g_r + m_r \geq 0\}. \quad (3)$$

Lemma 1 (Credible conditionality)**.** *In every subgame-perfect equilibrium, after alignment a: (i) the United States defends Europe in a crisis if and only if $s_a \geq K$; (ii) it denies frontier AI after exit if and only if $\nu + m_{ai} \geq k_X$, and after Atlantic alignment if and only if $\nu \geq k_A$; (iii) after exit it penalizes ASML if and only if $\varphi V \leq \gamma + m_f$, and it never penalizes ASML after A; (iv) China punishes Europe if and only if $a = A$ and $g_r + m_r \geq 0$. Relative to Atlantic alignment, exit therefore costs Europe*

$$c \equiv \kappa S + \chi_{ai}\,\alpha + \chi_f f + \eta, \quad (4)$$

and Atlantic alignment costs Europe $\chi_r r$ in Chinese punishment.

Part (ii) shows that the AI instrument is conditional only inside a scarcity window. When compute is plentiful ($\nu + m_{ai} < k_X$), the United States sells to Europe whatever its alignment. When compute is extremely scarce ($\nu \geq k_A$), it rations even a loyal Europe, as the scenario's 2029 Frontier Inference Services Rule does to most of the EU. In both cases denial does not depend on Europe's choice and drops out of c. Part (iii) is where the patron's dependence on the chokepoint enters: the more the United States needs ASML, the less credible a threat to cripple it. Jurisdiction over China-bound flows, captured by θ, has no such limit.

Substituting Lemma 1, Europe's payoffs are

$$u_E(A) = \tau_U - \chi_r r, \qquad u_E(X) = \lambda_C \tau_C - c, \quad (5)$$

up to terms common to both alignments, which affect Europe's welfare but not its choice.

Definition 1 (Exit premium)**.** Europe's *exit premium* is the most Europe could gain by exiting, net of the credible costs of exit, relative to Atlantic alignment with no US transfer:

$$Q \equiv \omega - c + \chi_r r = \lambda_C \min\{(1-\theta)W, B_C\} - \kappa S - \chi_{ai}\alpha - \chi_f f - \eta + \chi_r r. \tag{6}$$

The term follows **?**: a state's influence over a partner depends on what the partner would lose by leaving. In our setting the relevant quantity runs the other way, from the chokepoint holder's side. Q prices Europe's credible alternative, and the US dependence V appears in it only through χ_f.

4.3 Equilibrium: the exit premium sets the price

Proposition 1 (Leverage equals the exit premium)**.** *Suppose $Q \leq \min\{V, B_U\}$. As $\varepsilon \to 0$, every subgame-perfect equilibrium has Europe choosing A, with the US transfer converging to*

$$L \equiv [Q]^+ = \max\{0, Q\}. \tag{7}$$

Europe's payoff converges to $[Q]^+ - \chi_r r$, the United States' to $-[Q]^+$, and China's to zero. If instead $Q > \min\{V, B_U\}$, Europe exits.

We call L Europe's *leverage*: the transfer it extracts for keeping the chokepoint aligned. The logic is that of an auction in which the losing bidder sets the price. China can induce exit only by offering Europe more than the United States does, net of exit costs, and it will spend at most what exit is worth to it. The United States therefore pays exactly enough to make China's best credible offer unattractive. When $Q \leq 0$, even China's best offer cannot outweigh the costs the United States can credibly impose and the value Europe places on its alignment, so the United States pays nothing.

Three features of Proposition 1 drive the rest of the paper. First, V does not appear in L except through χ_f. A chokepoint the patron cannot do without is worth no more to its holder than one the patron barely needs, as long as the holder's credible alternative is the same. Second, the threshold $Q > 0$ is a credibility condition. It asks whether Europe would actually prefer China's best credible package to Atlantic alignment with nothing, once US responses are taken into account. Third, the condition $Q \leq B_U$ matters. If the United States cannot credibly promise enough, Europe exits even when keeping it aligned is worth more to Washington than exit is worth to Beijing.

4.4 Which results depend on who moves first

A bargaining theorist will ask whether V drops out only because the United States proposes first. It partly does. Proposition 2 collects the cases.

Proposition 2 (Protocol bounds)**.** *Assume ASML keeps supplying US-aligned fabs while bargaining continues (no hold-up), $V \geq Q$, and $B_U \geq V$. Let $L_0 \equiv [\chi_r r - c]^+$ and $\bar{L} \equiv \min\{V, [\lambda_C B_C - c + \chi_r r]^+\}$, the largest transfer a Chinese offer within its cap could force. Europe's leverage L is:*

(a) US offers first, China second: $L = [Q]^+$ (Proposition 1).

(b) Simultaneous offers: the Nash equilibrium set is $L \in [[Q]^+, \bar{L}]$; with undominated offers, $L = [Q]^+$.

(c) China offers first, the United States second: the subgame-perfect set is $L \in [L_0, \bar{L}]$; with undominated Chinese offers, $L \in [L_0, [Q]^+]$.

(d) Europe makes a take-it-or-leave-it demand of the United States, and China moves after a rejection: $L = V \cdot \mathbb{1}\{Q > 0\}$.

(e) Europe and the United States alternate offers, the status quo continues during bargaining, and Europe may exit after any rejection on the terms of China's best credible offer: in the stationary equilibrium $L \to [Q]^+$ as the discount factor approaches one.

*(f) Bargaining breaks down into exit with exogenous probability p per round (**?**): as $p \to 0$, $L \to [Q + \beta(V - Q)]^+$, where β is Europe's bargaining weight.*

Corollary 1 (A necessary condition)**.** *Under protocols (a)–(e), with undominated offers in (b) and (c), $Q \leq 0$ implies $L = 0$. The patron's dependence V raises Europe's leverage only through Europe's proposal power (d), through exogenous breakdown risk (f), or through hold-up, which we analyze in section 6.2.*

The robust content of the model, then, is the threshold. Whatever the bargaining protocol, a chokepoint holder with no credible alternative gets nothing for its chokepoint. How much of the patron's dependence it captures above the threshold depends on agenda power. Case (d) is the version of the argument that commentators have in mind when they call ASML Europe's last card: if Europe could commit to a demand, it could charge the United States the full value of the chokepoint. But a take-it-or-leave-it demand is itself a commitment, and the only threat that makes it bite is exit. Case (f) is Schelling's threat that leaves something to chance (**?**). If a dispute over ASML could slip out of control and end in realignment whether or not Europe's government wants it (say, through a change of government in The Hague), the United States must pay to avoid that risk even when $Q < 0$.

4.5 What the exit premium depends on

Proposition 3 links L to the three features of the European position this paper is about: defence dependence, the credibility of commitments, and economic dependence on China.

Proposition 3 (Determinants of leverage)**.** *Under the conditions of Proposition 1:*

(a) Defence dependence works only through conditionality. *L depends on S only through κS. If exit does not change the US stake in Europe's defence ($s_A = s_X$), then $\kappa = 0$ and no level of defence dependence gives the United States leverage over the chokepoint. Holding π_X fixed, L is nonincreasing in the reliability of the guarantee to a loyal Europe, π_A.*

(b) Compute scarcity creates and then destroys the AI lever. *As ν rises, Q falls by α when $\nu + m_{ai}$ crosses k_X, and rises by α when ν crosses k_A. Leverage is therefore non-monotone in the scarcity of compute.*

(c) Patron dependence protects the holder only against self-harming sanctions. *L is nondecreasing in V. It changes only when φV crosses $\gamma + m_f$ and firm-level penalties lose credibility, and then rises from $[Q]^+$ to $[Q + f]^+$. Jurisdiction over China-bound flows (θ) remains credible for every V.*

(d) Chinese coercion is a tax on the alliance. *The US transfer $[Q]^+$ rises one for one with $\chi_r r$ when $Q > 0$. Europe's payoff, $\max\{\omega - c, -\chi_r r\}$, is nonincreasing in r and nondecreasing in China's credible offer ω, so China's promises raise Europe's payoff only when $Q > 0$, and China's threats never do.*

(e) Unreliable US promises cause inefficient exit. *If $B_U < Q \leq V$, Europe exits even though the United States values Atlantic alignment more than China values exit.*

Part (a) turns on a distinction that alliance theory has long drawn between how much an ally depends on its patron and how much the patron can make that dependence contingent (**???**). Europe's need for US protection is enormous, but it becomes American leverage over ASML only if Washington would actually protect a loyal Europe more readily than a realigned one. The United States gets the most leverage from a guarantee that is reliable for loyal allies and void for disloyal ones, the arrangement **?** calls hierarchy. Two other kinds of guarantee give it none. An unconditional guarantee, one the United States would honour because it values European security for its own reasons, removes security from the bargain. So does a guarantee Europe no longer believes (π_A near π_X). The more doubt Washington casts on its commitment to allies who comply, the less leverage that commitment buys it over allies deciding whether to comply.

Part (b) captures a two-sided effect of the compute shortage the scenario describes. Scarcity makes denial cheap, which gives Washington a credible threat against an exiting Europe. Extreme scarcity leads Washington to ration loyal allies as well, which removes the threat's conditional part. The Frontier Inference Services Rule in the scenario, which halves allocations to a Europe that has done nothing, lowers the cost of exit at the same time as it lowers Europe's welfare.

Part (c) states precisely where the conventional view is right. The scenario observes that FDPR penalties on ASML "would also harm the US". The model agrees that this matters, but only for firm-level penalties, and only if the United States has no cheaper instrument aimed at China-bound flows. As section 2 documents, it does.

Part (d) separates the two channels through which Europe's economic dependence on China operates. China's capacity to punish Atlantic alignment raises what Washington must pay, but the payment only compensates Europe for the punishment it will absorb. China's capacity to reward exit raises Europe's payoff, but only when Europe already has leverage. De-risking that reduces China's coercive reach therefore never lowers Europe's payoff. De-risking that reduces the value of China's promises lowers it only in the region $Q > 0$, which, we argue in section 7, Europe has not

been in since 2019.

5 Control Rights and the Commitment Problem

The static model takes Europe's ownership of the chokepoint as given. The scenario's 2031 crisis is about ownership itself. Washington proposes to fold ASML into a joint holding company in which it holds a controlling vote over output, customers, and technology transfer. This section asks when a patron prefers to buy or coerce control of a chokepoint rather than pay rents to its holder, and what that does to the holder.

The answer turns on a commitment problem of the kind **?** and **?** identify in shifting-power settings, applied to an object whose ownership shapes future bargaining power (**?**). Europe's exit premium can change over time. Rearmament lowers κS; Chinese credit lines and market offers raise ω; years of being rationed under a tiered AI regime erode the conditional part of the AI lever (Proposition 3b). If Europe's premium is expected to turn positive, Washington faces future rents, and Europe cannot credibly promise to forgo them. Control rights remove the problem. Ownership determines who gets to choose alignment later, and the patron can buy it while its own threats still bite.

5.1 Setup

There are two periods, $t \in \{1, 2\}$, and a discount factor $\delta \in (0, 1]$. In each period in which Europe holds control, the static game of section 4 is played with period-specific parameters, giving exit premium Q_t and AI exposure α_t. We assume the AI lever is conditional in both periods ($\chi_{ai,t} = 1$) and that the United States wins each static contest ($V_t \geq Q_t + \alpha_t$, $B_U \geq V_t$).

At the start of period 1 the United States may issue an ultimatum: Europe transfers control over ASML's customers, output, and technology for both periods in exchange for a payment $T \geq 0$ made at signing; if Europe refuses, its access to frontier AI is cut in period 1. An ultimatum can be public, in which case failing to carry out the threat costs the United States an audience cost $m \in [0, \bar{m}]$ (**?**). If Europe accepts, it bears a sovereignty cost $\sigma \geq 0$, which includes the domestic political cost of surrendering a national champion and any capitalized loss of security from relocating capacity to US soil, and China punishes the transfer with $r^c \geq 0$. Once control has passed, the United States has no reason to cut Europe off: its ex post gain from doing so is negative, so the assurance half of the ultimatum is credible (**?**). If Europe refuses, the United States decides whether to cut. Cutting a Europe that remains aligned yields the direct ex post gain $g_N = \nu_1 - k_A$, which is negative outside the extreme-scarcity regime. It also removes the AI term from Europe's period-1 exit costs, so the period-1 exit premium rises from Q_1 to $Q_1 + \alpha_1$.

Payoffs, normalized as before, are:

Outcome	Europe	United States
No ultimatum	$[Q_1]^+ + \delta[Q_2]^+$	$-[Q_1]^+ - \delta[Q_2]^+$
Accept	$T - \sigma - r^c$	$-T$
Refuse, cut	$[Q_1 + \alpha_1]^+ - \alpha_1 + \delta[Q_2]^+$	$g_N - [Q_1 + \alpha_1]^+ - \delta[Q_2]^+$
Refuse, no cut	$[Q_1]^+ + \delta[Q_2]^+$	$-m - [Q_1]^+ - \delta[Q_2]^+$

Access is restored in period 2; Lemma 3 shows why.

5.2 The bite of a credible threat

Lemma 2 (Credible ultimatum and effective bite)**.** *After a refusal, the United States cuts Europe off if and only if*

$$m + g_N \;\geq\; [Q_1 + \alpha_1]^+ - [Q_1]^+. \tag{8}$$

When it does, Europe's net loss is the effective bite

$$b_1 \equiv \alpha_1 - \left([Q_1 + \alpha_1]^+ - [Q_1]^+\right) = \min\{\alpha_1, [-Q_1]^+\}. \tag{9}$$

Two features matter. First, without an audience cost the threat is not credible: with $m = 0$ and $g_N < 0$, cutting off a Europe that has merely refused to sell control costs Washington something and gains it nothing. That is why the scenario's White House has to go public. Second, a punishment hurts its target only up to the target's slack. Cutting AI access removes one of the reasons Europe stays aligned. Once $Q_1 + \alpha_1 > 0$, the rest of the cut comes back to Europe as rent, because Washington must now pay to keep it from exiting. A large stick adds nothing beyond the point where using it would make exit attractive, which is the logic of marginal deterrence (**?**).

Lemma 3 (Self-limiting sanctions)**.** *Extending the cut into period 2 costs the United States* $[Q_2 + \alpha_2]^+ - [Q_2]^+$ *in additional rent, which equals* α_2 *when* $Q_2 \geq 0$*. A permanent cut is credible only if the period-2 audience cost exceeds this amount less* g_N*.*

The scenario's threat that Europe will "lose all access, present and future, to American AI" is therefore not credible inside the model unless Washington has staked a great deal of reputation on it. A permanent cut hands Europe exactly the exit option the cut was meant to deter, at the moment Europe's future premium is positive.

5.3 Preventive control

Proposition 4 (Preventive control)**.** *Under complete information, the United States issues a public ultimatum, and Europe accepts, if and only if*

(i) Europe's future exit premium is positive, $Q_2 > 0$*;*

(ii) the threat can be made credible, $\bar{m} + g_N \geq [Q_1 + \alpha_1]^+ - [Q_1]^+$*; and*

(iii) the effective bite exceeds the deadweight cost of transferring control, $b_1 > \sigma + r^c$*.*

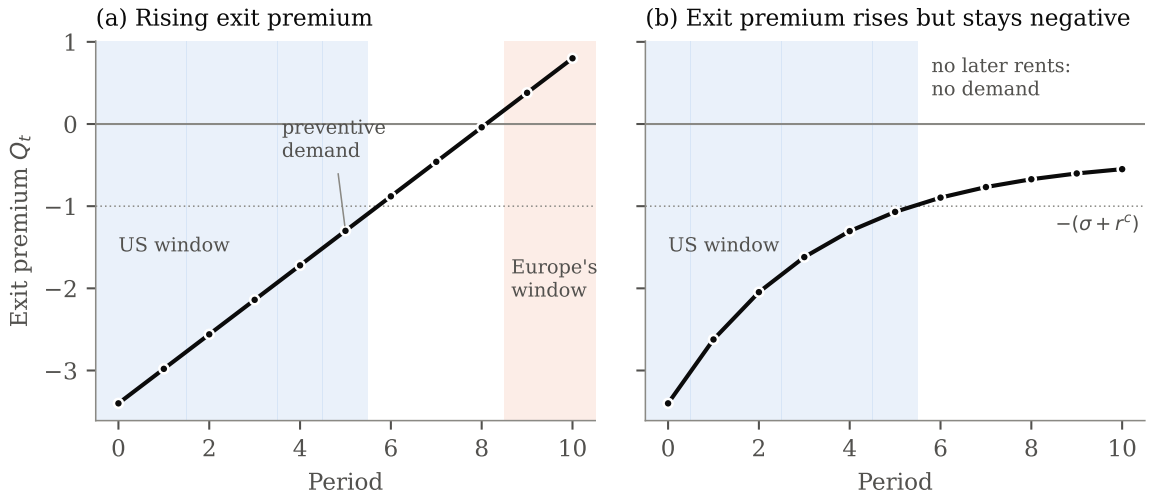


Figure 1: Windows for preventive control (illustrative units). Shaded blue: dates at which a credible US threat has a bite larger than $\sigma + r^c$, which requires $Q_t < -(\sigma + r^c)$. Shaded orange: Europe's window, $Q_t > 0$. In panel (a) the US window precedes Europe's, so Washington demands control at the last date of its window (Corollary 2). In panel (b) Europe's premium rises but never turns positive, there are no future rents to remove, and no demand is made.

The payment is $T^* = [\delta Q_2 + \sigma + r^c - b_1]^+$ *and Europe's payoff is* $\max\{\delta Q_2 - b_1,\ -\sigma - r^c\}$.

Because $b_1 \leq [-Q_1]^+$, condition (iii) requires $Q_1 < -(\sigma + r^c)$. Washington must act while Europe's exit premium is still well below zero, and it acts because that premium is expected to become positive. A patron that faces no future rents has no reason to seek control; a patron whose threats no longer bite cannot get it. Preventive control occupies the interval between those two states. The same logic extends to many periods.

Corollary 2 (Windows)**.** *In a T-period version in which the United States may issue an ultimatum at any date at which Europe still holds control, it does so at some date if and only if there is a date t at which the cut is credible and $b_t > \sigma + r^c$, and a later date $s > t$ with $Q_s > 0$.*

The US window, the set of dates at which (ii) and (iii) hold, and Europe's window, the dates with $Q_s > 0$, are separated by at least $\sigma + r^c$ in units of the exit premium. Under complete information the speed of the shift does not matter, only its direction and the existence of a window of credible coercion before Europe's window opens. Two features of Proposition 4 cut against intuition. China's threat to punish the transfer, r^c, protects European ownership, because it raises the bite Washington needs. And Europe's sovereignty costs, which commentators treat as an obstacle to a deal, play the same role: a Netherlands that would pay a high domestic price for surrendering ASML is harder to coerce into doing so.

5.4 Private sovereignty costs and breakdown

The scenario's endgame is a standoff, not a signature. Under complete information the ultimatum is either accepted or never issued. Breakdown requires that Washington be uncertain about what surrender would cost the European governments involved, which is plausible given coalition politics in The Hague and Berlin. Let σ be Europe's private information, uniformly distributed on $[0, \bar{\sigma}]$, and suppose conditions (i) and (ii) of Proposition 4 hold. Write $R \equiv \delta Q_2$, $a \equiv b_1 - r^c > 0$, and $e \equiv -g_N + [Q_1 + \alpha_1]^+ \geq 0$ for what it costs Washington to carry out the cut.

Proposition 5 (Breakdown)**.** *If the United States issues the ultimatum, Europe accepts if and only if $\sigma \leq s^*$, where*

$$s^* = \max\Big\{a - R,\ \min\big\{\bar{\sigma},\ \tfrac{a+e}{2}\big\}\Big\}. \tag{10}$$

With probability $P_b = [1 - s^*/\bar{\sigma}]^+$ Europe refuses, the cut is carried out, and Europe keeps ASML. P_b is nonincreasing in the bite b_1 and in the execution cost e, and nondecreasing in China's punishment of the transfer r^c and in $\bar{\sigma}$. When $a - R > \min\{\bar{\sigma}, (a+e)/2\}$, the United States offers nothing and P_b is increasing in Europe's future premium R.

Breakdown is the scenario's "neither" outcome: Europe refuses both great powers, loses frontier access, and keeps the chokepoint. The proposition says when it becomes likely. It is more likely when China credibly threatens to punish a transfer, when the European domestic cost of surrender is uncertain, and when Washington's threat is cheap to execute. The last point is the usual risk-return trade-off (**??**): a patron that loses little from a standoff makes tougher offers and is refused more often.

5.5 Promises or threats

Europe could try to remove the commitment problem itself. An allied export-control compact, dual-key licensing that gives Washington a veto over China-bound sales, or treaty-entrenched EU controls would bind Europe to Atlantic alignment in period 2 without transferring ownership. Suppose Europe can adopt such a compact unilaterally before period 1, at cost σ_c, and China punishes it with r^b.

Remark 1 (Self-binding)*.* When the conditions of Proposition 4 hold, Europe strictly prefers to bind itself if and only if $\sigma_c + r^b < \sigma + r^c$ and $\sigma_c + r^b < b_1 - \delta Q_2$. Binding removes the US motive for an ultimatum and keeps ownership in Europe, but it forfeits the future rents that a positive Q_2 would have earned.

A credible promise not to exit and a credible exit threat cannot coexist: the compact works by setting Europe's future exit premium to minus infinity. It does not remove the other source of leverage, the threat to withhold servicing from the United States itself, which section 6.2 analyzes. Binding against exit also removes the preventive part of Washington's stake, which can make the United States more willing to concede a hold-up dispute.

6 Extensions

The baseline treats Europe as one actor, gives it a single defection option, and lets the United States keep its chips flowing while the two sides bargain. Each extension relaxes one of these assumptions. The first asks who in Europe actually decides. The second gives Europe the power to hurt the United States during a dispute, which is where the patron's dependence on ASML returns. The third adds partial defection.

6.1 Who decides: nested chokepoints and veto players

Three features of the real decision structure matter. The Netherlands licenses ASML's exports to China, but a licence permits a shipment; it does not order one. ASML itself decides whether to ship, and its own exposure to US law, customers, suppliers, and shareholders is large. Turning ASML against the United States, finally, is not a Dutch decision at all: exports to the United States move under an EU general authorization, and the instrument designed for retaliation, the Anti-Coercion Instrument (ACI), requires qualified-majority decisions in the Council.

Let the firm's China profit from exit be y and its expected loss from US measures aimed at it be f^F. If $y < f^F$, the firm will not ship unless the state compels it at cost $k_s \geq 0$, for instance under emergency powers like those the Dutch government used against Nexperia in 2025. For the veto-player results, let each member state i have its own exit premium $Q_i = \omega_i - c_i + \chi_{r,i} r_i$, computed from its own share of China's offer and its own exposure to US and Chinese instruments. For the ACI, let q_i be member i's net gain from adopting a countermeasure against the United States, and let $\mathcal{B}$ be the family of blocking coalitions: at least four members holding more than 35 percent of EU population.

Proposition 6 (Nested chokepoints)**.** *(a)* Firm compliance. *Europe's effective exit premium is $Q - k_s \mathbb{1}\{y < f^F\}$. If the United States can raise f^F above y at negligible cost, Europe's leverage is at most $[Q - k_s]^+$.*

(b) Holder veto. *If exit requires the consent of member h alone, the United States pays $[Q_h]^+$. A conditional benefit granted to h alone and withdrawn upon exit raises c_h one for one and lowers this payment.*

(c) Blocking minorities. *If no one bids for adoption, the United States defeats an ACI countermeasure at cost $K(0) = \min_{S \in \mathcal{B}} \sum_{i \in S} [q_i]^+$. If a rival bidder values adoption at $B > 0$ and moves second, the United States must make every qualified majority cost at least B; its cost $K(B) = \min_{\tau \geq 0} \sum_i \tau_i$ subject to $\sum_{i \in S} [\tau_i - q_i]^+ \geq B$ for every qualified majority S can exceed $K(0)$ by more than B and generally requires payments to members outside any minimal blocking coalition (**?**).*

Part (a) is the institutional point in **?** turned around. A state can weaponize a network hub only if its domestic institutions give it control over the hub. ASML meets the network condition but sits inside networks the United States controls: US-origin subsystems, US persons, and US customers.

Europe's chokepoint is nested in American ones, and the exit threat is only as credible as the Dutch state's willingness to compel its own champion to break US law. Part (b) is divide and rule. Granting the Netherlands, and only the Netherlands, preferential AI access, as the scenario's tier system does (the real January 2025 AI Diffusion Rule similarly split the EU, placing ten members in its top tier for reasons unrelated to ASML), gives the veto player more to lose from exit than its neighbours. Part (c) corrects a piece of arithmetic. Under the 2026 Council population figures (**?**), the states the scenario says opposed or abstained on the 2029 countermeasure (the Netherlands, Ireland, Poland, the three Baltic states, and Italy) hold 28.05 percent of EU population, short of a blocking minority, and any blocking coalition that excludes Germany, France, Italy, and Spain needs at least twelve members. The pivotal members on a measure against the United States are large states, not the eastern flank.

6.2 Voice through hold-up: time to pain

The baseline assumes that bargaining takes place under the status quo, with ASML servicing US-aligned fabs throughout. The scenario's 2031 standoff is different. Once Europe refuses the ultimatum, Washington cuts off frontier AI, and Europe could interrupt servicing, spare parts, and tool deliveries in return. The scenario claims that Washington "has amassed enough EUV machines and maintenance know-how to forego ASML access for longer than Europe can forego access to frontier AI". That is a claim about time to pain, and the patron's dependence on the chokepoint enters through it.

Let Europe and the United States dispute a divisible issue: the scope of US control over ASML, or the terms of Europe's AI access. An agreement leaves Europe a share $p \in [0,1]$, with payoffs $-(1-p)s_E$ to Europe and $(1-p)s_U$ to the United States. Until they agree, Europe bears a flow cost c_E from date τ_E on, and the United States bears c_U from date τ_U on. The buffers are what each side can live on before the cutoff hurts. For the United States, that means stockpiled tools, spare parts, and transferred maintenance know-how. For Europe, it means compute on European soil and stored model weights. Offers arrive at intervals Δ; the United States proposes with probability ψ and Europe otherwise. We take $\Delta \to 0$ and assume Europe's exit option is unattractive. Write $\hat{c}_i \equiv c_i/s_i$ and $D \equiv \tau_U - \tau_E$.

Proposition 7 (Time to pain)**.** *Suppose $\psi\hat{c}_E \neq (1-\psi)\hat{c}_U$.*

(a) The equilibrium is unique and agreement is immediate. If Europe is the less vulnerable side once both buffers are exhausted, $\psi\hat{c}_E < (1-\psi)\hat{c}_U$, its share is $p^* = 1$ when $D \leq 0$ and $p^* = [1-\psi\hat{c}_E D]^+$ when $D > 0$. If instead $\psi\hat{c}_E > (1-\psi)\hat{c}_U$, then $p^* = 0$ when $D \geq 0$ and $p^* = \min\{1, (1-\psi)\hat{c}_U|D|\}$ when $D < 0$.

(b) Let $\tau_E = 0$ and suppose the United States bears no cost unless Europe interrupts servicing. Interruption, which raises Europe's flow cost to $c_E + \ell$ and imposes c_U on the United States after τ_U, is played in a Markov perfect equilibrium if and only if $(1-\psi)\hat{c}_U > \psi(c_E+\ell)/s_E$ and $\tau_U < H_E \equiv s_E/[\psi(c_E+\ell)]$. Europe's share is then $1 - \tau_U/H_E$.

The proposition separates the two quantities **?** called sensitivity and vulnerability. The buffer gap D measures sensitivity: who feels the cutoff first. The post-buffer ranking of normalized costs measures vulnerability: who could hold out longer once neither has a cushion. Hold-up gives Europe leverage only if it wins on vulnerability and the US buffer is shorter than Europe's endurance horizon H_E. In the interior, each unit of US buffer costs Europe ψc_E, Europe's own rate of pain weighted by US agenda power; the US cost c_U matters only for who wins on vulnerability. This is where V re-enters the analysis. The more US frontier compute depends on continued ASML servicing, the larger c_U. The scenario's claim that a stockpile lets Washington outlast Europe amounts to the claim that $\tau_U \geq H_E$. Its Epilogue's call for compute on European soil amounts to raising τ_E. Servicing and spare parts matter more than tools here: installed EUV machines keep running, but they need engineers and consumable parts, so the US buffer for servicing is short even if its stock of tools is large.

6.3 Hedging and marginal deterrence

The scenario's 2028 crisis, like the real disputes of 2023–2025, was not about exit. It was about whether ASML could keep selling and servicing older immersion tools in China. Add a third alignment, hedging (H), with China's credible offer $\omega_H < \omega$ and exit costs c_H, and suppose the United States, after observing a hedge, may withhold an instrument k that also deters exit. Let D_k be the instrument's contribution to Europe's exit costs (κS for security, α for AI) and $g_k(H)$ the United States' ex post gain from withholding it from a hedging Europe.

Proposition 8 (Marginal deterrence)**.** *Withholding instrument k to punish hedging is credible if and only if*

$$m_k + g_k(H) \;\geq\; [Q + D_k]^+ - [Q]^+. \tag{11}$$

Enforcing jurisdiction over the hedged flows themselves leaves Q unchanged and is credible whenever its direct cost is small. If $Q + D_k > 0$ and audience costs are small, the United States cannot use security or AI linkage against hedging, and must rely on flow-specific jurisdiction.

A patron cannot spend the sanctions that deter full defection on partial defection without weakening deterrence of the full version (**??**). The prediction is that Washington polices legacy-tool sales with targeted jurisdiction over those sales (de minimis rules for lithography tools bound for China, the foreign direct product rule, servicing restrictions) and keeps security and AI linkage in reserve. This matches the scenario's 2028 crisis, where the threat is the FDPR, and it matches the actual US sequence from 2019 to 2024. Jurisdictional threats of this kind also work under uncertainty about US resolve. If the firm-level penalty f_H is existential for ASML while the revenue at stake is a fraction of its sales, even a small probability that Washington means it is enough. That is the formal content of the scenario's line that ASML "cannot afford to call Washington's bluff".

7 Europe 2031 and the Record, Revisited

We use the model in two ways. First we ask which of the scenario's episodes are equilibria of the game it describes. Then we ask whether the record since 2019 looks like what the model predicts for a holder whose exit premium is negative. Neither exercise is a test. The parameters are not measured, and a negative exit premium can rationalize almost any act of compliance after the fact (**?**). The model earns its keep by making predictions about *which* instruments are used, *when* demands for control arise, and *who* is pivotal, and those can be checked against the record.

7.1 How much leverage? A back-of-envelope

Figure 2 gives a sense of magnitudes. Set every exit cost except security to zero, which is generous to Europe, so that the exit premium is $\omega - \kappa S$. Europe has leverage only if the conditionality of the US guarantee is below a break-even value $\kappa^* = \omega/S$. Take Bruegel and Kiel's estimate of EUR 250 billion a year as the cost of replacing US protection, and value China's credible offer at ASML's 2025 sales to China, about EUR 9.5 billion. Then κ^* is about 0.04. Europe would have leverage only if it believed that realigning would raise the probability of being left undefended by less than four percentage points. Even if China would pay ASML's entire annual revenue for Europe's realignment, the threshold rises only to about 0.13. Adding the AI lever, firm-level penalties, and Europe's own aversion to strengthening China's AI capabilities lowers every line in the figure. The calculation is crude: the model's quantities are flows and probabilities, not budget lines. But it makes the central point concrete. The value of the chokepoint to the rival is an order of magnitude smaller than the value of the guarantee the patron can make conditional.

7.2 The scenario

Table 1 summarizes what the model says about each episode.

2028. The scenario's first crisis is an equilibrium of the game it describes. The demand is to stop hedging, not to prevent exit, and Proposition 8 predicts that Washington will use flow-specific jurisdiction rather than the security or AI levers it needs to deter exit. That is what the scenario's Washington does when it threatens the FDPR. Because the penalty would be existential for the firm and the revenue at stake is a fraction of its sales, compliance follows even if the threat is partly a bluff. Europe gets nothing because its hedging premium is negative, not because it negotiated badly. The scenario's remedy for 2028, more solidarity among member states, would not have changed this: solidarity does not reduce US jurisdiction over a firm whose light source is made in San Diego.

2029. The Frontier Inference Services Rule has an effect the scenario does not draw out. By cutting allocations to most of Europe while it remains aligned, Washington removes the conditional part of its AI lever. In the model this raises Europe's exit premium (Proposition 3b) and starts the drift that motivates the 2031 ultimatum. The failed countermeasure is harder to square with the

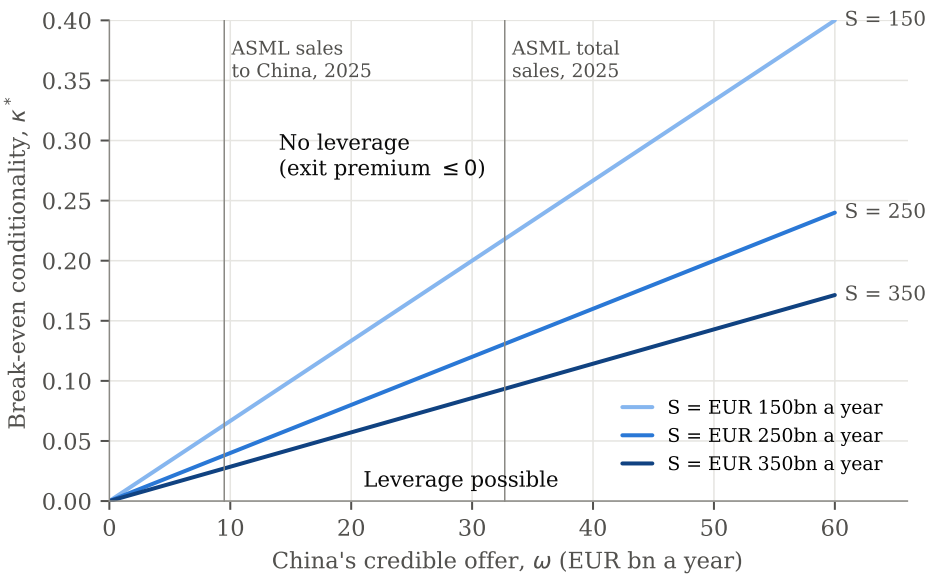


Figure 2: Break-even conditionality of the US security guarantee. Lines show $\kappa^* = \omega/S$ for three values of the annual cost S of replacing US protection. Above a line Europe's exit premium is negative and its leverage is zero. All other exit costs are set to zero, which favours Europe. Vertical rules mark ASML's 2025 sales to customers in China and its total 2025 sales (**?**).

model, because it is harder to square with the Council's voting rules. Under the 2026 population weights, the Netherlands, Ireland, Poland, the three Baltic states, and an abstaining Italy hold 28.05 percent of the EU's population, so twenty members with almost 72 percent remain in favour, more than the 55 percent of members and 65 percent of population that a qualified majority requires. For the measure to fail, Germany, France, or Spain would have to defect as well. The pivotal members on any use of ASML against Washington are large states, which are exactly the members a US divide-and-rule strategy would need to target.

2031. The ultimatum is a preventive demand for control rights of the kind Proposition 4 characterizes. Europe is drifting: Chinese credit lines raise China's credible offer, five years of being treated as a vassal erode the value Europe places on a guarantee to the loyal, and rationing has removed the AI lever's conditional part. A forward-looking Washington acts while its threat still bites, and it goes public to make the threat credible. Two features of the scenario do not survive. The first is the threat that Europe will lose "all access, present and future" to American AI. Lemma 3 shows that a permanent cut hands Europe the exit option at the moment its future premium is positive, so the threat is credible only for a limited period. The second is the scenario's suggestion that China's counter-threat leaves Europe worse off whatever it does. China's threat is conditioned on signing with Washington. By Propositions 4 and 5, a threat of that kind raises the bite Washington needs and the probability that the ultimatum is refused, which protects European ownership.

Table 1: Europe 2031 through the model

Episode	What the model says	Result used
2028: the Dutch halt DUV servicing in China	An equilibrium. The demand concerns hedging, so Washington uses jurisdiction over the hedged flows (the FDPR), not linkage. With high US jurisdictional reach and existential firm-level penalties, Europe's hedging premium is negative and it receives nothing. Solidarity from other member states does not change the reach of US jurisdiction.	Propositions 1 and 8
2029: tiered AI access and the failed countermeasure	Rationing a loyal Europe removes the conditional part of the AI lever and raises Europe's exit premium. The countermeasure's failure does not follow: the named opponents hold 28.05 percent of EU population. Tier 1 status for the Netherlands alone is divide and rule.	Proposition 3(b), Proposition 6
2031: the ultimatum	A preventive demand for control rights, issued as Europe's exit premium is expected to turn positive and made public to make the threat credible. The threat of permanent exclusion is not credible. China's threat to punish signing raises the bite Washington needs and the chance of breakdown.	Propositions 4 and 5, Lemma 3
2031: the US stockpile	Plausible for tools, much less so for servicing and parts. It matters only if the US buffer exceeds Europe's endurance horizon.	Proposition 7
Epilogue: compute on European soil	Raises Europe's buffer and makes AI denial costly for Washington, which lowers the conditional AI lever.	Proposition 3(b), Proposition 7
Epilogue: a coalition of middle powers	Japan and Korea face the same structure. A coalition helps only if its assets are complements for the rival and its members can compensate one another; otherwise Washington buys the cheapest member.	Proposition 6

7.3 The record, 2019–2026

Compliance with in-kind payment. The Dutch sequence from 2019 to 2025 is what Proposition 1 predicts for a holder in the region $Q \leq 0$: alignment without a cash transfer. What the Netherlands did receive fits the model's account of in-kind transfers. The December 2024 rule exempted countries with equivalent controls from the new foreign direct product rules (**?**). That exemption costs Washington almost nothing, keeps the formal decision in The Hague, and can be withdrawn at will. It is a carrot made of the same jurisdiction that Proposition 3(c) identifies as Washington's cheapest stick.

Servicing and the MATCH Act. By 2026 the live dispute was servicing of China's installed base, the hedge in our terms, and the instrument Washington reached for was flow-specific jurisdiction backed by legislation (**?**), as Proposition 8 predicts. A statute also does something the model treats as an audience cost: it ties the hands of the executive and makes the threat harder to bargain away. The Dutch response was voice without exit. The prime minister raised the bill with the president and the trade minister lobbied Congress (**?**), but no one in The Hague suggested that the Netherlands might stop aligning. Two other features of 2026 also fit the model. Beijing's April 2026 regulations against foreign extraterritorial measures raised the firm's cost of complying with US servicing bans (**?**). Chinese customers pulled orders forward to stockpile tools, which is the wasting-asset problem in concrete form (**?**).

Nexperia. The Nexperia dispute shows China's coercive capacity at work and where its costs fell. China punished a Dutch move that followed a US rule by withholding chips from European carmakers, which spread the cost across member states. The Dutch government suspended its order, and relief came mainly through the US–China deal at Busan, in which Washington suspended its own affiliates rule (**??**). In the terms of Proposition 3(d), China's coercion was a tax on the alliance, and the United States ended up paying it.

The guarantee. The model's most direct implication for the transatlantic relationship concerns conditionality. Washington's 2025 strategy document links "favorable treatment" to export-control alignment (**?**), which raises conditionality. Troop withdrawals and open doubt about the commitment to loyal allies lower the value of the guarantee to the loyal, π_A, which lowers it. By May 2026 only 11 percent of respondents in a fifteen-country poll saw the United States as an ally (**?**). If Europeans come to believe that Washington will not defend them whatever they do, defence dependence stops buying Washington leverage over ASML. The Dutch trade minister's complaint about the MATCH Act is consistent with that change, though one episode cannot establish it.

8 Implications and Limitations

8.1 What would make ASML a card worth holding

The model identifies which policies would raise Europe's leverage and which would not. Several cut against positions common in the European debate.

Reduce conditional exposure, not dependence as such. Rearmament raises Europe's leverage over ASML only insofar as it lowers κS, the part of Europe's security that Washington could withhold after realignment. Capabilities that Washington could not withdraw selectively, and an independent European deterrent above all, matter more for leverage than aggregate spending targets. By the same logic, a United States that values European security for its own sake, and would defend Europe whatever it did with ASML, would have less leverage over the chokepoint, not more.

Shrink the nesting, at a price. Europe's exit threat is limited by US jurisdiction over ASML's subsystems, staff, and customers (Proposition 6a). Moving the EUV light source and US-origin components out of US jurisdiction would raise the exit premium. It would also be costly, slow, and read in Washington as preparation for exit, which is the trigger for a preventive demand (Proposition 4).

Treat servicing as the lever. Europe's one source of leverage that does not run through China is the power to interrupt servicing during a standoff (Proposition 7). It works only if Europe can outlast the US buffer, which makes compute on European soil and stored model weights relevant. It also requires a commitment device, because interruption is costly to Europe, and EU procedures are slow where US instruments act immediately.

Expect a preventive demand if Europe's premium rises. Any policy that succeeds in raising Europe's exit premium also raises Washington's incentive to demand control while its threats

still bite. A Europe that anticipates this can bind itself to Atlantic alignment through an allied export-control compact that keeps ownership in European hands (remark 1). Doing so gives up the exit threat in exchange for keeping the asset. The choice between a credible promise and a credible threat is the central strategic decision the model puts to European governments.

De-risk from China's threats. Reducing China's coercive reach over European supply chains never lowers Europe's payoff (Proposition 3d). Reducing the value of China's promises lowers it only when Europe already has leverage, which on our reading of fig. 2 it has not had.

For Washington the lessons are symmetric. Its leverage over ASML depends on a guarantee that is reliable for loyal allies, and every signal that it is not erodes that leverage. Rationing loyal allies' access to frontier AI destroys the AI lever. Firm-level penalties against ASML are a weak threat because carrying them out would damage America's own supply. A preventive grab for control requires a public commitment and risks a standoff, while an allied compact would secure the same alignment at lower cost.

8.2 Limitations

The limitation that should most worry a hostile reader is measurement. Every result is stated in terms of quantities, such as conditionality, China's credible offer, and the buffers in a standoff, that are hard to observe, and a negative exit premium can rationalize compliance after the fact. We have tried to answer this with predictions about instruments, timing, and pivots rather than outcomes, and with the order-of-magnitude comparison in fig. 2. A sharper test would measure how European elites assess the conditionality of the US guarantee, for instance in a survey experiment that varies whether US protection is described as contingent on export-control alignment, and relate those assessments to support for resisting US export demands.

Three other choices limit the analysis. The great powers are unitary. Congress, the courts, and US firms that sell to China shape the credibility of American threats, and the MATCH Act shows Congress acting as a commitment device. Most results assume complete information. Uncertainty about Washington's resolve, about Europe's sovereignty costs, or about China's progress would produce standoffs more often than the baseline predicts, as Proposition 5 illustrates. Finally, the stakes of the AI race are exogenous. If frontier AI conferred a decisive and lasting advantage, the values of the chokepoint to both great powers would rise sharply and the preventive logic of section 5 would apply with more force. The model does not say whether that premise is correct.

9 Conclusion

Europe 2031 calls ASML Europe's last card. The model in this paper suggests that the card is real but that its value is set by a quantity Europe has never held in abundance: a credible alternative to Atlantic alignment. A chokepoint gives its holder leverage over the patron that depends on it only when the rival's credible offer exceeds the costs the patron can credibly impose, plus the value the holder itself places on alignment. For a Europe that depends on Washington for its defence and for

frontier AI, that condition has not held since the Netherlands first declined to license an EUV sale to China in 2019. The patron's dependence on the chokepoint enters only through sanctions that would hurt the patron, the holder's agenda power, the risk of breakdown, and the holder's capacity to outlast the patron in a standoff over servicing.

The analysis connects to a broader debate about weaponized interdependence. Control of a hub is a source of power only when domestic institutions let the hub's host use it and when the host has somewhere else to go. ASML fails the first condition because its own supply chain runs through American jurisdiction, and Europe fails the second because its security does. Where those conditions change, the model predicts that the patron will try to secure control of the hub before its own threats lose their bite. In the scenario that moment comes in March 2031. In the record, the December 2024 exemption and the 2026 fight over servicing restrictions show Washington tightening its jurisdiction while The Hague still has no exit.

References

A Proofs

Throughout, $[x]^+ = \max\{0, x\}$. Payoffs are normalized as in section 4.1.

Proof of Lemma 1

Stage (4) decisions are taken after Europe's alignment is fixed and after transfers have been paid, so each decision maximizes the decision maker's ex post payoff. (i) In a crisis with cost K, defending yields the United States $s_a - K$ and not defending yields zero, so it defends iff $s_a \geq K$, an event of probability $G(s_a) = \pi_a$. (ii) After exit, denial yields $\nu - k_X$ plus avoidance of the audience cost m_{ai} attached to the public denial rule; supplying yields zero. Denial occurs iff $\nu + m_{ai} \geq k_X$. After Atlantic alignment no commitment is at stake and denial occurs iff $\nu \geq k_A$. Because $k_A > k_X$ and $m_{ai} \geq 0$, denial after A implies denial after X, so the conditional part of the instrument is $\chi_{ai} = \mathbb{1}\{\nu + m_{ai} \geq k_X\}\mathbb{1}\{\nu < k_A\}$. (iii) After exit, penalizing ASML yields $\gamma - \varphi V$ and failing to do so costs m_f; the United States penalizes iff $\varphi V \leq \gamma + m_f$. After Atlantic alignment, penalizing yields $-\varphi V < 0$ and no commitment is at stake. (iv) After A, punishment yields China g_r and avoids audience cost m_r; it punishes iff $g_r + m_r \geq 0$. After X China has obtained what it sought and punishment has no rationale. Europe's expected security loss after a is $(1 - \pi_a)S$, so exit adds $(\pi_A - \pi_X)S = \kappa S$. The other terms follow directly, and terms common to both alignments (for instance, AI denial when $\nu \geq k_A$) cancel. □

Proof of Proposition 1

Fix the grid mesh $\varepsilon > 0$ and assume B_U and B_C are grid points. By Lemma 1 and eq. (5), Europe chooses A iff $\tau_U - \chi_r r \geq \lambda_C \tau_C - c$.

China's best response. Given τ_U, China wins only with an offer τ_C satisfying $\lambda_C\tau_C > \tau_U - \chi_r r + c$. Let $\underline{\tau}_C(\tau_U)$ be the smallest grid point with this property. Winning yields $(1-\theta)W - \underline{\tau}_C(\tau_U)$, losing yields zero, and China does not bid at zero profit. China therefore wins iff $\underline{\tau}_C(\tau_U) \le B_C$ and $\underline{\tau}_C(\tau_U) < (1-\theta)W$. Since $\underline{\tau}_C(\tau_U) \in \big((\tau_U - \chi_r r + c)/\lambda_C,\ (\tau_U - \chi_r r + c)/\lambda_C + \varepsilon\big]$, China wins whenever $\tau_U < Q - \lambda_C\varepsilon$ and loses whenever $\tau_U \ge Q$.

The US offer. Any $\tau_U \ge [Q]^+$ that is a grid point keeps Europe aligned. The United States' cheapest such offer, τ_U^ε, satisfies $[Q]^+ - \lambda_C\varepsilon \le \tau_U^\varepsilon \le [Q]^+ + \varepsilon$ (the lower bound because offers in $[Q - \lambda_C\varepsilon, Q)$ may also deter China, depending on where the grid falls). Keeping Europe aligned at this price is better for the United States than losing V iff $\tau_U^\varepsilon \le V$, and feasible iff $\tau_U^\varepsilon \le B_U$. Under $Q \le \min\{V, B_U\}$ this holds in the limit, so every subgame-perfect equilibrium has $\tau_U^\varepsilon \to [Q]^+$ and Europe choosing A. Payoffs follow from eq. (5). If $Q > \min\{V, B_U\}$, for small ε the United States either cannot or will not offer Q, and China wins. □

Proof of Proposition 2

Write $\bar{L} \equiv \min\{V, [\lambda_C B_C - c + \chi_r r]^+\}$, the largest US transfer a Chinese offer within the cap can force. Under the stated assumptions, $Q \le \bar{L} \le V$.

(a) This is Proposition 1.

(b) In any Nash equilibrium in which Europe chooses A, the US offer must satisfy $\tau_U \ge [Q]^+$; otherwise China could deviate to a profitable winning offer. The US offer must also equal $[\lambda_C\tau_C - c + \chi_r r]^+$ for China's actual offer τ_C; otherwise the United States could lower its offer and still win. Any $\tau_C \le B_C$ with $\lambda_C\tau_C - c + \chi_r r \in [[Q]^+, \bar{L}]$ supports such an equilibrium, with the United States offering exactly that amount: China loses, earns zero, and cannot profitably win because $\tau_U \ge [Q]^+$; the United States prefers matching to losing because the required transfer is at most V. China's offers above $(1-\theta)W$ are weakly dominated, because if accepted they yield a loss. With undominated offers, $\lambda_C\tau_C - c + \chi_r r \le Q$, so $L = [Q]^+$.

(c) Given τ_C, the United States offers $[\lambda_C\tau_C - c + \chi_r r]^+$ if that is at most V, and otherwise lets Europe exit. Because $Q \le V$, China cannot win at a price below its valuation, so China earns zero and is indifferent among all losing offers. Every $\tau_C \in [0, B_C]$ with $\lambda_C\tau_C - c + \chi_r r \le V$ is part of a subgame-perfect equilibrium, which gives $L \in [L_0, \bar{L}]$, with $L_0 = [\chi_r r - c]^+$ obtained at $\tau_C = 0$. Excluding weakly dominated offers above $(1-\theta)W$ gives $L \in [L_0, [Q]^+]$.

(d) After a US rejection, China can induce exit at a profit iff $Q > 0$, by offering slightly more than $(c - \chi_r r)/\lambda_C$. If $Q > 0$ the United States accepts any demand up to V, so Europe demands V. If $Q \le 0$ a rejection leaves Europe aligned with no transfer, so the United States rejects every positive demand.

(e) The status quo gives the United States the alignment it wants and gives Europe nothing, so absent exit the United States accepts no positive transfer and the inside-option payoff is zero. Exit yields Europe $\omega - c$ per period against $-\chi_r r$ under the status quo. Suppose $Q > 0$. In the stationary equilibrium the United States, when it proposes, offers the per-period transfer Q, the least that stops Europe from rejecting and exiting. When Europe proposes, the United States

accepts a transfer τ forever iff $-\tau/(1-\delta) \geq -\delta Q/(1-\delta)$, so Europe demands δQ. Both converge to Q as $\delta \to 1$. If $Q \leq 0$, exit is not a credible response to a rejection and the United States accepts no positive transfer. This is the outside-option principle (**??**).

(f) With exogenous breakdown into exit, the limit of the alternating-offers game as $p \to 0$ is the Nash bargaining solution with breakdown payoffs as the disagreement point (**?**): $-V$ for the United States and $\omega - c$ for Europe. Maximizing $(V-\tau)^{1-\beta}(\tau - \chi_r r - \omega + c)^{\beta}$ gives $\tau = Q + \beta(V-Q)$, truncated at zero. □

Proof of Proposition 3

(a) By eq. (6), S enters only through κS, and $\kappa = G(s_A) - G(s_X) = 0$ when $s_A = s_X$. Holding π_X fixed, $\partial Q/\partial \pi_A = -S \leq 0$, and $[Q]^+$ is nondecreasing in Q. (b) χ_{ai} switches from 0 to 1 at $\nu = k_X - m_{ai}$ and from 1 to 0 at $\nu = k_A$, with $k_X - m_{ai} < k_A$. (c) V enters Q only through $\chi_f = \mathbb{1}\{\varphi V \leq \gamma + m_f\}$, which is nonincreasing in V; when it switches from 1 to 0, Q rises by f. (d) By Proposition 1 Europe's payoff is $[Q]^+ - \chi_r r = \max\{-\chi_r r, \omega - c\}$. (e) See the last step of the proof of Proposition 1. □

Proof of Lemmas 2 and 3

After a refusal, cutting yields the United States $g_N - [Q_1 + \alpha_1]^+ - \delta[Q_2]^+$ and not cutting yields $-m - [Q_1]^+ - \delta[Q_2]^+$; comparing gives eq. (8). Europe's payoff falls by $\alpha_1 - ([Q_1+\alpha_1]^+ - [Q_1]^+)$. If $Q_1 \geq 0$ this is zero; if $-\alpha_1 < Q_1 < 0$ it is $-Q_1$; if $Q_1 \leq -\alpha_1$ it is α_1. Hence $b_1 = \min\{\alpha_1, [-Q_1]^+\}$. The period-2 statement follows by the same comparison applied to period 2. □

Proof of Proposition 4

Suppose (ii) holds, so the United States can choose m large enough to make the cut credible. Europe's refusal payoff is $\rho_E = [Q_1]^+ - b_1 + \delta[Q_2]^+$, and it accepts iff $T - \sigma - r^c \geq \rho_E$. The cheapest acceptable payment is $T^* = [\rho_E + \sigma + r^c]^+$. The United States issues the ultimatum iff $-T^* > -[Q_1]^+ - \delta[Q_2]^+$, i.e., iff its gain $\Gamma \equiv [Q_1]^+ + \delta[Q_2]^+ - T^*$ is positive.

If $b_1 = 0$, then $T^* \geq [Q_1]^+ + \delta[Q_2]^+$ and $\Gamma \leq 0$. If $b_1 > 0$, then $Q_1 < 0$, so $\Gamma = \delta[Q_2]^+ - [\delta[Q_2]^+ - b_1 + \sigma + r^c]^+ = \min\{\delta[Q_2]^+, b_1 - \sigma - r^c\}$, which is positive iff $Q_2 > 0$ and $b_1 > \sigma + r^c$. If (ii) fails, refusal costs Europe nothing, $T^* = [Q_1]^+ + \delta[Q_2]^+ + \sigma + r^c$, and $\Gamma \leq 0$. Europe's payoff when the ultimatum is issued is $T^* - \sigma - r^c = \max\{\delta Q_2 - b_1, -\sigma - r^c\}$. □

Proof of Corollary 2

Call a date t *eligible* if the cut is credible at t, $b_t > \sigma + r^c$, and $Q_s > 0$ for some $s > t$. In any subgame starting at a date with no later eligible date, the argument in the proof of Proposition 4, with $\delta[Q_2]^+$ replaced by the discounted sum of later rents, shows that no ultimatum is issued after that date. Let t^* be the last eligible date. In the subgame beginning at t^* with Europe in control, the continuation after a refusal contains strictly positive rents and no further ultimatum, so the

United States' gain from issuing is $\min\{\text{discounted future rents}, b_{t^*} - \sigma - r^c\} > 0$, and it issues. On the equilibrium path an ultimatum is therefore issued at t^* or earlier. Conversely, if no date is eligible, then by backward induction no subgame contains an ultimatum: at each date either the cut is not credible, or the bite is too small, or there are no later rents to remove, and in each case the gain from issuing is nonpositive. We also verified the statement numerically on 100,000 random parameter draws with up to six periods (appendix B). □

Proof of Proposition 5

With conditions (i)–(ii) of Proposition 4, the United States sets $m \geq e$, so it carries out the cut after a refusal. Europe accepts iff $T - \sigma - r^c \geq -b_1 + R$, i.e., iff $\sigma \leq s \equiv T + a - R$. Relative to issuing no ultimatum, the United States gains $R - T$ if Europe accepts and loses e if Europe refuses. Write $x = R - T \leq R$, so $s = a - x$. For $s \in (0, \bar{\sigma})$ its expected gain is

$$\Pi(x) = \frac{(a - x)\, x - (\bar{\sigma} - a + x)\, e}{\bar{\sigma}},$$

which is strictly concave with unconstrained maximizer $x^\circ = (a - e)/2$, so $s^\circ = (a + e)/2$. If $s \geq \bar{\sigma}$, every type accepts and $\Pi = x$, which is increasing in x, so the United States never sets $s > \bar{\sigma}$. Imposing $s \leq \bar{\sigma}$ and $x \leq R$ gives $s^* = \max\{a - R, \min\{\bar{\sigma}, (a + e)/2\}\}$. The comparative statics follow because s^* is nondecreasing in $a = b_1 - r^c$ and in e, and $P_b = [1 - s^*/\bar{\sigma}]^+$. In the corner $s^* = a - R$, s^* is decreasing in R. The United States issues the ultimatum iff $\Pi(x^*) > 0$; in the interior this is $(a + e)^2 > 4\bar{\sigma}e$, and with $e = 0$ it reduces to $a > 0$. □

Proof of Remark 1

If the conditions of Proposition 4 hold, Europe's payoff without binding is $\max\{\delta Q_2 - b_1, -\sigma - r^c\}$. With binding, Europe has no future exit premium, the United States gains nothing from an ultimatum, and Europe's payoff is $-\sigma_c - r^b$ (period-1 rents are zero because $Q_1 < 0$). Binding is strictly better iff $-\sigma_c - r^b$ exceeds both terms. □

Proof of Proposition 6

(a) If $y < f^F$ the firm refuses to ship after a licence, and exit occurs only if the state compels it, which costs k_s; substituting into Proposition 1 gives the result. (b) China needs only h's consent, so the argument of Proposition 1 applies to h's payoffs. A benefit Δ that h receives under A and loses under exit raises c_h by Δ. (c) With no rival bidder, a countermeasure is adopted iff the members with $q_i > \tau_i$ form a qualified majority, which the United States prevents most cheaply by paying $[q_i]^+$ to every member of some blocking coalition. With a rival bidder moving second, the rival buys a qualified majority S at cost $\sum_{i \in S}[\tau_i - q_i]^+$ whenever that is less than B, which gives the linear program. Example: three members, adoption by two, $q_i = 0$. Then $K(0) = 0$, while $K(B)$ requires $\tau_i + \tau_j \geq B$ for every pair, so $K(B) = 3B/2$ with $\tau_i = B/2$ for all three. □

Proof of Proposition 7

(a) Let $w(t)$ be Europe's expected share at date t before the proposer is drawn. A proposer offers the responder exactly its continuation value, and agreement is immediate because delay is weakly costly to both. Hence

$$w(t) = \psi \max\{w(t+\Delta) - \hat{c}_E(t)\Delta,\, 0\} + (1-\psi)\min\{1,\, w(t+\Delta) + \hat{c}_U(t)\Delta\}.$$

After both buffers are exhausted the costs are constant; in the interior $w(t) - w(t+\Delta) = [(1-\psi)\hat{c}_U - \psi\hat{c}_E]\Delta$, so in an infinite horizon the only bounded solution is $w = 1$ if $(1-\psi)\hat{c}_U > \psi\hat{c}_E$ and $w = 0$ otherwise (cf. the fixed-cost case of **?**). Between the two buffer dates only one side bears costs. If $D > 0$, Europe alone bears costs on $[\tau_E, \tau_U)$ and w falls (moving forward in time) at rate $\psi\hat{c}_E$, truncated at zero; if $D < 0$, the United States alone bears costs on $[\tau_U, \tau_E)$ and w rises at rate $(1-\psi)\hat{c}_U$, truncated at one. Before the earlier buffer date nobody bears costs and w is constant. Taking $\Delta \to 0$ gives the stated shares. (b) Without interruption the United States bears no cost and Europe bears c_E from date zero, so $p^* = 0$. With interruption, part (a) with $\tau_E = 0$ gives $p^* = [1 - \psi\tau_U(c_E + \ell)/s_E]^+ = [1 - \tau_U/H_E]^+$ if Europe is the less vulnerable side post-buffer, and $p^* = 0$ otherwise. Interruption is played iff it raises p^*. □

Proof of Proposition 8

After a hedge, the United States compares withholding k, which yields $g_k(H)$ but removes k from Europe's exit costs, so that the subsequent static contest costs it $[Q + D_k]^+$, with not withholding, which costs m_k if it had committed to punish hedging and leaves the contest at $[Q]^+$. Jurisdiction over hedged flows does not enter Europe's exit costs, so enforcing it leaves the contest at $[Q]^+$. □

B Numerical verification

We checked each result by brute force in Python, with scripts in the replication folder. Proposition 1 was checked by backward induction on a grid of offers (mesh 0.01) over random parameter draws with $V \geq Q$. Proposition 4 was checked on 200,000 random draws with ties never broken in favour of the threat, and Corollary 2 on 100,000 draws with two to six periods; both produced zero mismatches. Proposition 5 was checked against direct numerical optimization of the US offer. Proposition 7 was checked by backward induction in discrete time with $\Delta = 0.001$; deviations from the closed form are $O(\Delta)$ once the stationary phase is long enough to reach its boundary value.